

NASA LaRC & NASA GRC

Lawrence Giron Jr.



Summer 2023

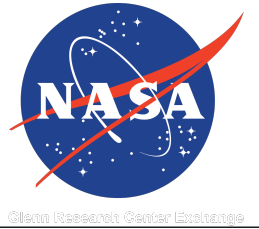
NASA Langley Research Center
Hampton, Virginia

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Fall 2023

NASA Glenn Research Center
Cleveland, Ohio

About Me



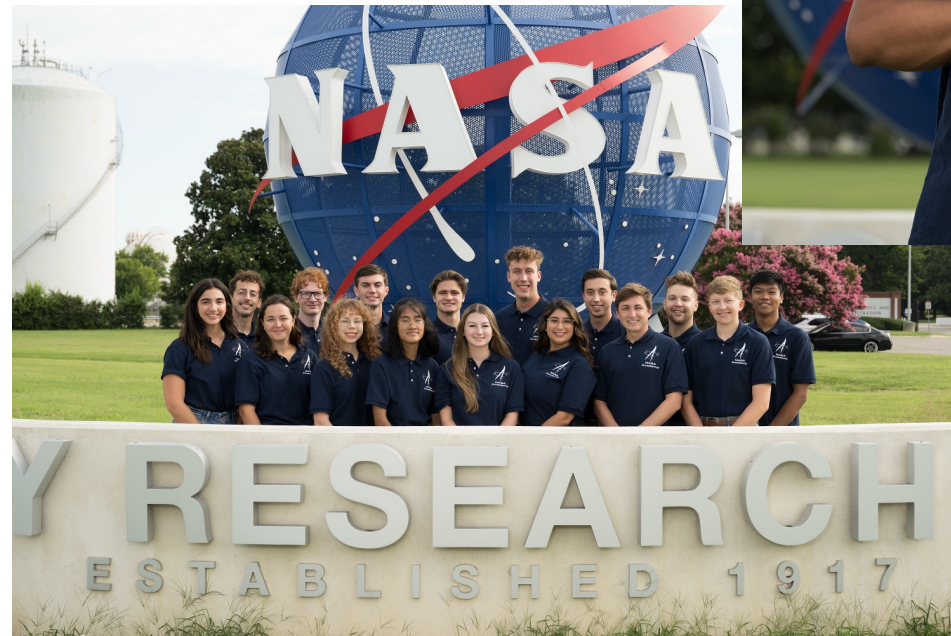
Lawrence Giron Jr.

- BS Mechanical Engineering – Spring 2023
 - ME Research Assistant
 - Engineering Academy Teaching Assistant
- NASA LaRC Academy – Summer 2023
 - Research Associate
- NASA GRC Intern – Fall 2023
 - Intelligent Control and Autonomy Intern
- MS Mechanical Engineering Student – Current
 - Atmospheric Corrosion Research with Dr. Srinivasan



NASA Academy Aeronautics Research Directorate In-Time System-Wide Safety Assurance

- 10 Week Program
- Multi-Disciplinary Research Project
- 16 Students from around the country
- Solve real problems in Aeronautics and Aerospace using available resources on center
- Technical Advisory Committee composed of NASA experts



Multiple Research Projects

- CLASSy – Compact Lightweight Aerial Sensor System
- Ad-Hawk – Aerial Connectivity Network
- VTOL HALE – Vertical Take-off and Landing High Altitude Long Endurance Vehicle
- NextNav – Beacon Based GPS System



Compact Lightweight Aerial Sensor System (CLASSy)

Kyleigh Anderson*[£]
West Virginia University, Morgantown, WV

Chad Brander*
Virginia Polytechnic Institute and State University, Blacksburg, VA

Eloisa Carrasco[£]
University of Minnesota Twin Cities, Minneapolis, MN

Reagan Courville[†]
Louisiana State University, Baton Rouge, LA

Adriana de la Guardia^{££}
The College of William and Mary, Williamsburg, VA

Jasmine Fensler*
Embry-Riddle Aeronautical University, Daytona Beach, FL

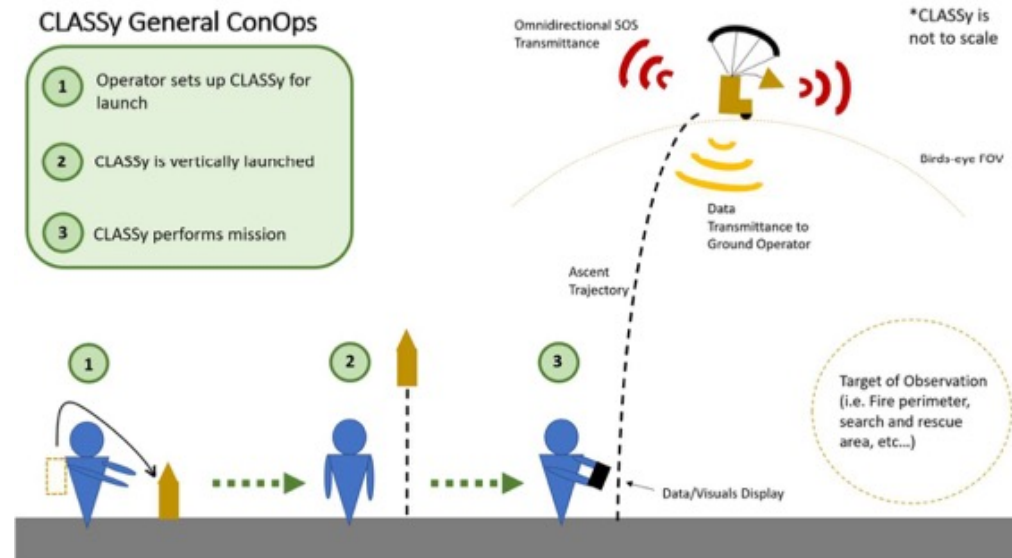
Lawrence Giron Jr.*^{££}
University of Alaska Anchorage, Anchorage, AK

John Oberlies*[£]
University of Michigan, Ann Arbor, MI

William Pankiewicz*
University of Kansas, Lawrence, KS

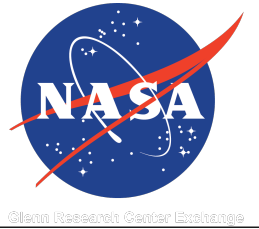
Grace Wachter*
University of Kansas, Lawrence, KA

- Designed and tested a cost-effective disaster mitigation tool that provides an aerial perimeter point of view for first responders in natural disaster situations
- Studied, designed, and tested parachute recovery systems to provide stable and steady descents of launched sensors



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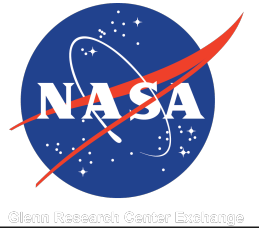


- Able to use the resources and facilities on center for project testing
- Gantry for rocket descent system testing
 - CERTAIN Range for UAS Testing



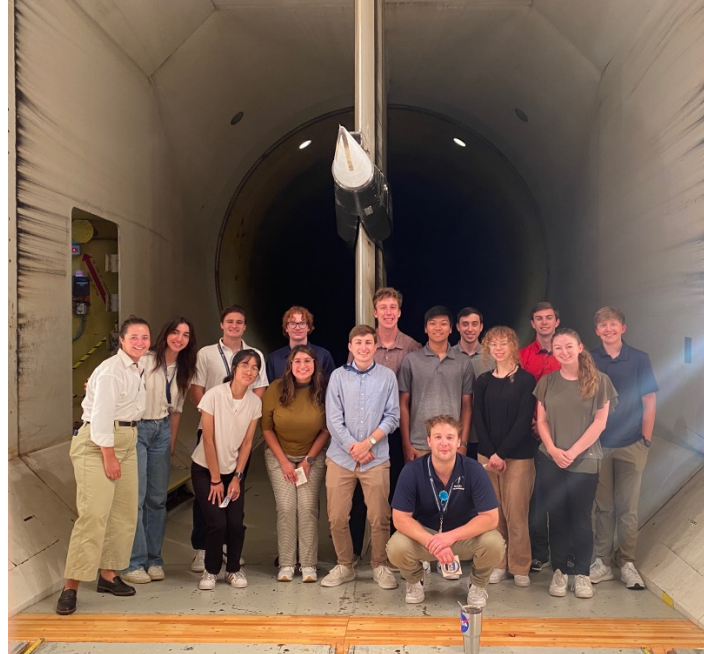
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Projects and Tours include

- F-22 Tour at the Military Base
- Multiple Wind Tunnels across the center
- Wallops Flight Facility at Wallops Island, VA
- Gantry
- Materials Testing Lab
- Dragonfly Project
- X-59 Quesst
- Material testing for the Mars sample recovery mission
- Many More



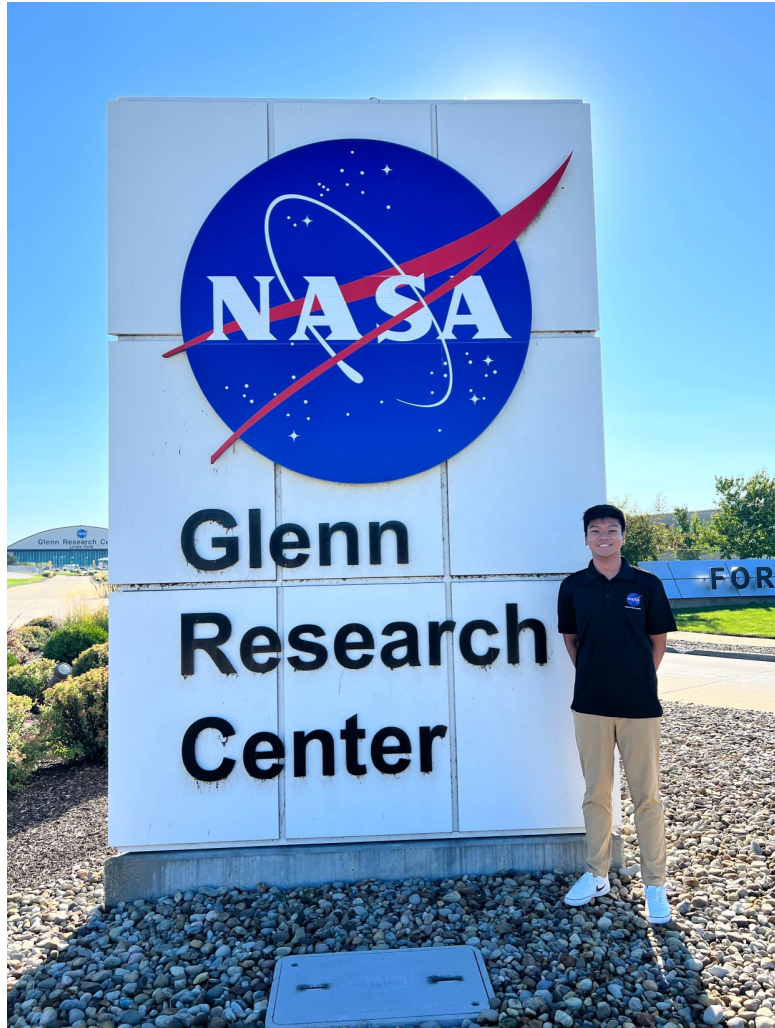
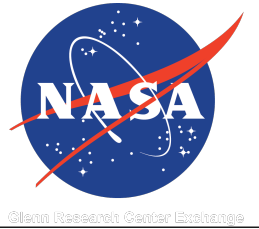


Take Aways from NASA LaRC

- Not only learned about engineering concepts and principles
- Learned how to work in a multidisciplinary team of engineers from different backgrounds
- Gained experience turning to professional engineers around the center to gain input and knowledge about the problems we were facing in our projects
- Gain experience with research and writing a technical research paper leading to a publication on the NASA Technical Reports Server

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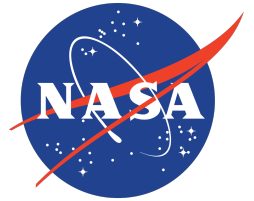
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Aeronautics Research Mission Directorate
Intelligent Control and Autonomy

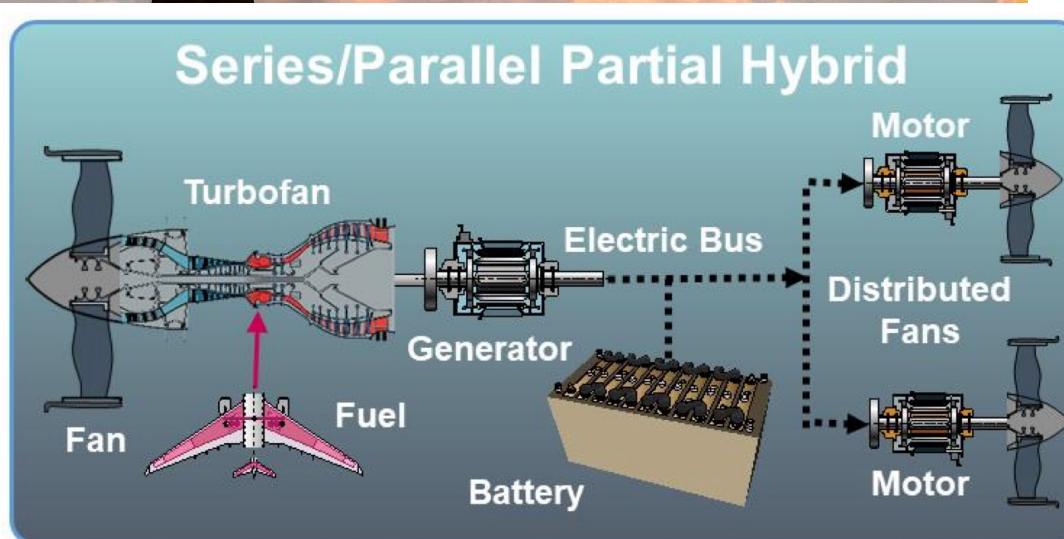
SUSAN – Subsonic Single Aft
Engine Hybrid-Electric
Passenger Aircraft Concept
Model

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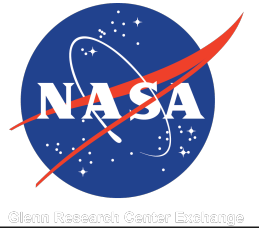
Glenn Research Center Exchange



- Expanded the SUSAN powertrain model for the optimization and management of thermal health
- Thermally modeled the electrical components of the aircraft simulation model with the MATLAB Simulink Electrical Modeling and Thermal Analysis Toolbox (EMTAT) developed by NASA
- Make additions to the EMTAT toolbox
- 2023 Energy & Mobility Conference and Expo

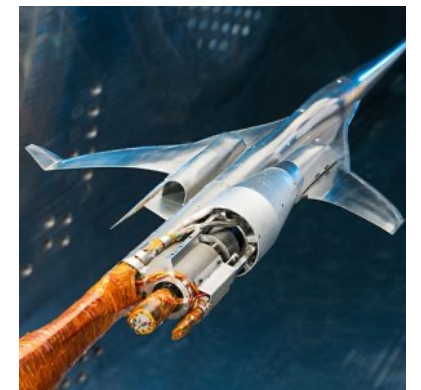
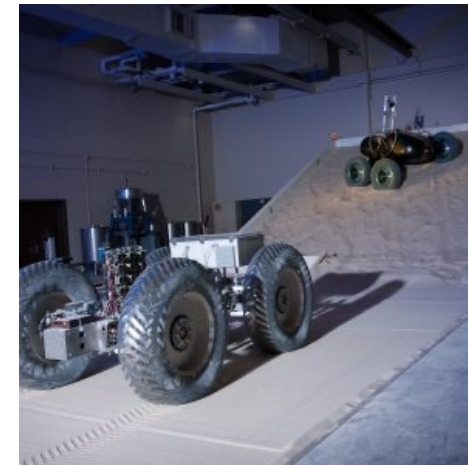
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Projects and Tours Include

- Multiple Subsonic and Supersonic Wind Tunnels across the center
- Microgravity Research Facility – Drop Tower
- Planetary Exploration Test Facility – SLOPE Lab
- Electric Propulsion and Power Laboratory
- Neil Armstrong Test Facility in Sandusky, OH



NASA GRC Space Apps Hackathon Competition

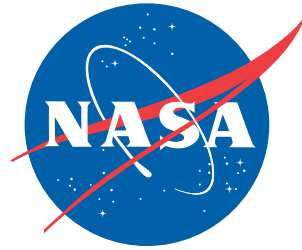
- Project Awarded First Place
- Developed VULCAN Fire Response Ops – Virtual Utility for Locating, Containing, and Assisting Notification designed to enhance fire prevention, monitoring, and response operations using machine learning, SMS notifications, and available data from NASA satellites





Take Aways from NASA GRC

- Was able to directly apply what I learned during my BSME at UAA
- Gained experience working on a full-scale research project spanning the different research centers across the country across many different branches within the project
- Gained experience working alongside NASA engineers and ensure my work was accurate
- Was able to present my work to the branch and prepare it for an updated public release in the future



Lawrence Giron Jr.
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